



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

PHYSICS
PAPER 4 Practical Test

9188/4

INSTRUCTIONS TO SUPERVISORS

NOVEMBER 2015 SESSION

2 hours 30 minutes

Great care should be taken that any confidential information given does not reach the candidates either directly or indirectly.

The Supervisor may have access to the Question Paper a working day before the examination.

When preparations are over, the Question Paper is to be re-sealed and locked up with the other copies of the Question Paper as soon as possible.

These Supervisor's Instructions consists of 6 printed pages and 2 blank pages.

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Instructions for preparing apparatus

1. The apparatus required for each question is set out on pages 3 and 4. Enough apparatus must be provided to ensure that
 - (a) no candidate is kept waiting at the end of the allocated time for each question,
 - (b) spare apparatus is immediately available in the event of failure or breakage of any item.
2. It is assumed that ordinary apparatus of a Physics laboratory will be available.

Instructions for the Practical Physics Supervisor

1. These instructions are enough for the Supervisor to make the right decisions.
2. Candidates should be informed that if they find themselves in real difficulty they may ask the Supervisor for practical assistance but that the extent of this assistance will be reported to the Examiner who may make a deduction of marks.
3. The Supervisor should complete the information required on page 7 and 8 of these instructions and enclose those pages in the envelope containing the answers of the candidates. A note of any help given to or any particular difficulties experienced by a candidate should also be enclosed, especially if the Examiner would be unable to discover these from the written answers.
4. The SUPPLIER has been notified of the apparatus required. Should the Supervisor decide to improvise some or part of the apparatus, he/she must ensure that this **does not** make the examination more or less accessible to the candidates. The Supervisor should give details to the examiner if the apparatus used differ(s) in any significant way from that specified in these Instructions.
5. It is assumed that candidates will provide themselves with such standard items as a digital stopwatch, a 30 cm rule, a pair of compasses, 0° to 180° protractor and set squares. Squared paper should be available.
6. Whenever a stopwatch or stop clock is specified, candidates should be advised, in advance that they may, if they wish, use quartz wristwatches with stopwatch facilities.
7. It is assumed that the number of sets of apparatus for any one experiment will be at least half the number of candidates. Where necessary, Supervisors should direct candidates as to which experiments they are to perform first. The experiments carry the same mark value.
8. **The supervisor MUST perform the experiments and submit with the candidates' answers a sample set of results in the same form as that expected of the candidates.**

1. Apparatus per candidate.

Each candidate to be supplied with

- (i) a metre rule (1 mm) x 1
- (ii) half metre rule x 3
- (iii) 60 cm strings x 4
- (iv) Stop watch (0.01s or better) x 1
- (v) Stand, boss and clamp x 2 each

N.B The supervisor should set up Fig.1.1 for the candidate.

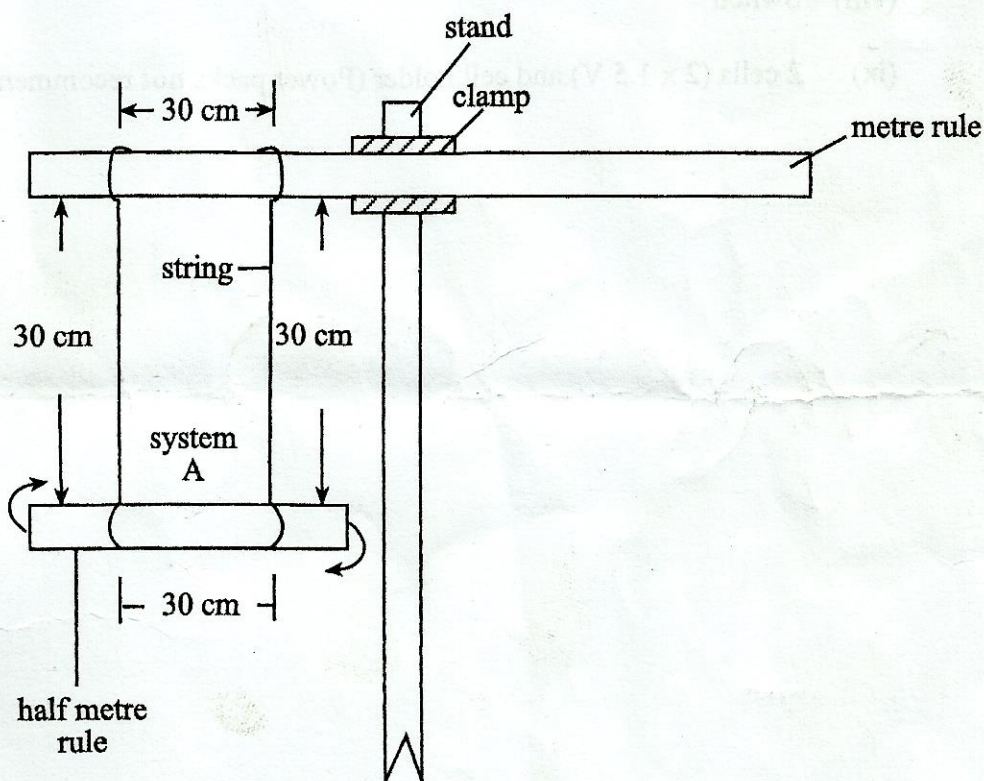


Fig.1.1

2. Apparatus per candidate

- (i) micrometer screw gauge
- (ii) voltmeter f.s.d 5 V and (0.01 V)
- (iii) ammeter f.s.d ≥ 3 A (0.01 A)
*It has been found that digital metres are more suitable
- (iv) crocodile clip ✓
- (v) jockey (easy to use for the candidates)
- (vi) 7 connecting leads ✓
- (vii) 1.00 m long 34 swg constantan wire which can be fixed to a meter rule. Wire must be labelled X.
- (viii) Switch
- (ix) 2 cells (2 x 1.5 V) and cell holder (Power packs **not** recommended).